

Lecture 2:

1. What is an Operating System?

Operating System is software that helps you use your computer and hardware.

2. What is a Kernel?

Kernel is the core part of the OS or as you put it "the heart and brain of the computer".

3. Which Other Parts Aside From the Kernel Identify an OS?

- **Shell:** The interface where you type commands.
- **User Space:** Where user applications and processes run.

4. What is Linux and Linux Distribution?

- **Linux:** An open-source operating system kernel that forms the foundation of many distributions.
- **Linux Distribution:** A complete OS built around the Linux kernel.

6. List at Least 4 Linux Characteristics:

- **Open Source:** Anyone can view, modify, and share it.
- **Multitasking:** Runs multiple processes at once without crashing.
- **Secure:** Strong security features.
- **Portable:** Works on almost any device.

7. What is Debian?

Debian is an old and reliable Linux distribution. It's community-driven and is the base for many other distributions like Ubuntu.

9. List and Define the Different Types of Licensing Agreements

- **Proprietary License:** You can't modify or share the software; you just pay to use it.
- **Open Source License:** You can view, modify, and share the software.

10. What is Free Software? Define the 4 Freedoms

Free Software means you can use, modify, and share it however you want. The 4 freedoms are:

- **Freedom 0:** Use it for any purpose.
- **Freedom 1:** examine the source code and modify it as you see fit.
- **Freedom 2:** Share it with others.
- **Freedom 3:** Share your modified version.

11. What is Virtualization?

Virtualization allows you to run multiple virtual machines or environments on one physical machine."A computer inside a computer".